

KEN1540 Exam

KEN1540 Numerical Mathematics 22/23 (period 5-resit semester 2) · 8 exercises · 80.0 points

1 Computer Arithmetic and Algebraic Equations

12.0 points · 4 questions

Consider the polynomial $p(x) = 1.03x^4 - 5.34x^2 - 0.0156x + 2.78$. Direct evaluation of $p(x)$ at $x = 1.51$ using 3-digit rounded arithmetic gives the value -4.08.

Text

a Evaluate $p(x)$ at $x = 1.51$ in nested (Horner) form using 3-digit rounded arithmetic. Give a detailed answer.

5.0 points · Open · 9/10 Page

+1 point

Write $p(x)$ in nested (Horner) form: $((((1.03x + 0)x - 5.34)x - 0.0156)x + 2.78$.

+3 points

Evaluate $p(x)$ in 3-digit rounded arithmetic.

+1 point

Correct answer: -4.05 (3 sf).

b Compute the absolute and relative error for direct evaluation **and** evaluation using nested (Horner) form at $x = 1.51$, given that the exact value is -4.05446831 (8 dp). Which method performs better?

3.0 points · Open · 47/100 Page

+0.5 points

Absolute error for direct evaluation.

+0.5 points

Absolute error for nested form evaluation.

+0.5 points

Relative error for direct evaluation.

+0.5 points

Relative error for nested form evaluation.

+1 point

Nested evaluation performs better than direct evaluation.

-0.5 points

Errors should be rounded to 2 sf.

Perform two steps of bisection to find the root of $p(x)$ in $[1.5, 2.5]$. Then, compute the residual of the estimate in the second step.

3.0 points · Open · 17/20 Page

+1 point

First step of bisection.

+1 point

Second step of bisection.

+1 point

Residual is $|f(2.125)| = |-0.354| = 0.354$.

d How many steps of bisection would you need if you want to find the root in $[1.5, 2.5]$ up to an accuracy of 10^{-3} . (You do **not** have to compute any bisection steps for this subquestion.)

1.0 point · Open · 1/2 Page

+1 point

At least 9 iterations.

2 Differential Equations

12.0 points · 3 questions

Consider the initial value problem $y' = \cos(ty) - t^2$ with $y(0) = 0.0000$.

Text

a Use a predictor-corrector method to find the solution for $t \in [0, 1.5]$ and $h = 0.5$. Use the two-stage Adams--Bashforth method for the prediction step and **the most appropriate** Adams--Moulton method for the correction step. Bootstrap your calculations using values of $w_1 \approx y(0.5)$ for **the most appropriate** of the methods given in the table below. Motivate your choices for the Adams-Moulton method and the bootstrapping.

Euler's method	$w_1 = 0.50000000$
2nd-order Runge-Kutta method	$w_1 = 0.45602090$
3rd-order Runge-Kutta method	$w_1 = 0.45614919$
4th-order Runge-Kutta method	$w_1 = 0.45545299$

9.0 points · Open · 1 3/4 Page

+1 point

Bootstrap using RK method of same order.

+1 point

That means order 3 because AB2+AM2 is order 3.

+1 point

Use same stage, i.e., AM2 because AB2.

+1 point

AB2 formula

+1 point

AM2 formula

+1 point

w2 prediction

+1 point

w2 correction

+1 point

w3 prediction

+1 point

w3 correction

-1 points

incorrect rounding

b Calculate the absolute and relative error of $y(1.5)$, given the exact value is 0.2400 3580 (8 dp).

1.0 point · Open · 1/2 Page

+0.5 points

Relative error.

+0.5 points

Absolute error

c Roughly, what would you expect the absolute error to be if you were to use $h = 0.25$ instead?

2.0 points · Open · 9/20 Page

+1 point

$O(h^3)$

+1 point

Factor 8

3 Polynomial Interpolation

12.0 points · 3 questions

Consider the function f on $[0, 3]$ given by $f(x) = \frac{1}{1+x}$.

The n th-order derivative of $f(x)$ is given by $f^{(n)}(x) = \frac{(-1)^n n!}{(x+1)^{n+1}}$.

Text

a Find the **cubic** Taylor polynomial $T_3(x)$ around $x_0 = 1$ for $f(x)$.

3.0 points · Open · 3/4 Page

+1 point

Taylor polynomial formula.

+1 point

Correct evaluation of derivatives.

+1 point

Final expression is $T_3(x) = 0.5000 - 0.2500(x - 1) + 0.1250(x - 1)^2 - 0.0625(x - 1)^3$.

b Use divided differences to compute the **cubic** polynomial $P_3(x)$ interpolating $f(x)$ at the values $x_0 = 0$, $x_1 = 1$, $x_2 = 2$, and $x_3 = 3$. You may assume that $f[x_0, x_1] = -0.5000$, $f[x_1, x_2] = -0.1667$, $f[x_2, x_3] = -0.0833$, $f[x_0, x_1, x_2] = 0.1667$, and $f[x_1, x_2, x_3] = 0.0417$.

5.0 points · Open · 9/10 Page

+1 point

Divided differences formula.

+2 points

Divided differences table + evaluation of $f[x_i]$ and correct placement of given values.

+1 point

Computation of $f[x_0, x_1, x_2, x_3]$.

+1 point

Final expression is $p_3(x) = 1.0000 + x(-0.5000 + (x - 1)(0.1667 + (x - 2)(-0.0417)))$.

c Estimate the value of $f(x)$ at $x = 1.5$ using $T_3(x)$ and $P_3(x)$. Determine the absolute error for both cases. What can you say about the errors and why? What would you expect to happen at $x = 2.5$?

4.0 points · Open · 9/10 Page

+1 point

Evaluation of $T_3(1.5) = 0.3984$

+1 point

Evaluation of $P_3(1.5) = 0.3906$

+1 point

Absolute errors are $1.6\text{e-}3$ and $9.4\text{e-}3$

+1 point

Taylor provides a good approximation close to the expansion point. Newton attempts a good approximation over the whole interval.

4 Numerical Integration & Differentiation

12.0 points · 2 questions

Consider the definite integral $\int_1^3 \sin(x^2) dx$.

Text

a Compute the Romberg estimate $R_{3,3}$ using four decimal places for the computed values. You may assume that $R_{2,0} = 0.3074$ and $R_{3,0} = 0.4281$.

10.0 points · Open · 1 4/5 Page

+1 point

Romberg formula

+1 point

Correct use of formula

+1 point

Trapezoid formula

+1 point

$R_{0,0} = T_1$

+1 point

$R_{1,0} = T_2$

+2 points

$R_{1,1}, R_{2,1}, R_{3,1}$

+2 points

$R_{2,2}, R_{3,2}, R_{3,3}$

+1 point

Correct answer 0.4685 (4 dp)

-1 points

Not using 4 dp or incorrect use.

b Give an error estimate for $R_{3,2}$.

2.0 points · Open · 1/2 Page

+2 points

Error estimate for $R_{3,2}$ is $\frac{R_{3,1} - R_{2,1}}{15} = 1.0 \cdot 10^{-2}$.

5 Least-Squares Approximation

12.0 points · 4 questions

The least-squares approximation to a function f over $[-1, +1]$ is given by

$$q_n(x) = \sum_{k=0}^n c_k P_k(x)$$

where the P_k are the Legendre polynomials.

Text

Suppose the c_k are given by

\$\$\$ \begin{array}{c|c|c|c|c|c} k & 0 & 1 & 2 & 3 & 4 & 5 \\ \hline

c_k & 0.3086 & 0.2193 & -0.0842 & -0.1233 & -0.0267 & 0.0262 \end{array} \$\$\$

Text

a Use the recurrence relation to evaluate $P_k(0.7)$ for $k = 0, 1, \dots, 4$.

4.0 points · Open · 7/10 Page

+1 point

Recurrence relation $P_k(x) = (2 - 1/k)xP_{k-1}(x) - (1 - 1/k)P_{k-2}(x)$.

+1 point

$P_0(0.7) = 1.0000$ and $P_1(0.7) = 0.7000$.

+1 point

$P_2(0.7) = (2 - 1/2) \times 0.7 \times 0.7000 - (1 - 1/2) \times 1.0000 = 0.7350 - 0.5000 = 0.2350$.

+1 point

$P_3(0.7) = (2 - 1/3) \times 0.7 \times 0.2350 - (1 - 1/3) \times 0.7000 = 0.2742 - 0.4667 = -0.1925$.

$P_4(0.7) = (2 - 1/4) \times 0.7 \times -0.1925 - (1 - 1/4) \times 0.2350 = -0.2358 - 0.1763 = -0.4121$.

0 points

Maybe change so $P_3(0.7)$ and $P_4(0.7)$ are 1 point each, and $P_0(x), P_1(x)$ combine with $P_k(x)$.

b Compute $q_4(0.7)$.

2.0 points · Open · 3/5 Page

+1 point

$q_4(0.7) = \sum_{k=0}^4 c_k P_k(0.7) = 0.3086 \times 1.0000 + 0.2193 \times 0.7000 - 0.0842 \times 0.2350 - 0.1233 \times -0.1925 - 0.0267 \times -0.4121$.

+1 point

$q_4(0.7) = 0.3086 + 0.1535 - 0.0198 + 0.0237 + 0.0110 = 0.4771$. (At most 4dp.)

Explain how orthogonality of the P_k gives rise to a simple formula for the square integral

$$\int_{-1}^{+1} q_n(x)^2 dx.$$

3.0 points · Open · 3/5 Page

+1 point

The integral can be written as a sum of terms involving $P_i(x) \times P_j(x)$ as

$$\sum_{i,j=1}^n c_i c_j \int_{-1}^{+1} P_i(x) P_j(x) dx.$$

+1 point

When $i \neq j$, the terms $\int_{-1}^{+1} P_i(x) P_j(x) dx = 0$ by orthogonality, so only terms involving $\int_{-1}^{+1} P_i(x) P_i(x) dx$ remain.

+1 point

The integral is then $\sum_{k=0}^n c_k^2 \int_{-1}^{+1} P_k(x)^2 dx$.

OPTIONAL

$\sum_{k=0}^n \alpha_k c_k^2$ where $\alpha_k = \int_{-1}^{+1} P_k(x)^2 dx = 2/(2k+1) = 1/(k + \frac{1}{2})$ (if α_k not in 5d).

d Compute the square integral $\int_{-1}^{+1} q_2(x) dx$.

3.0 points · Open · 7/10 Page

0 points

Mistake in question, should be $\int_{-1}^{+1} q_2(x)^2 dx$.

+1 point

Using $\alpha_k = \int_{-1}^{+1} P_k(x)^2 dx = 1/(k + \frac{1}{2})$, we have $\int_{-1}^{+1} q_2(x)^2 dx = \sum_{k=0}^2 \alpha_k c_k^2$.

OR

$$\int_{-1}^{+1} q_2(x) dx = \sum_{k=0}^2 c_k \int_{-1}^{+1} P_k(x) dx.$$

+1 point

$$\int_{-1}^{+1} q_2(x)^2 dx = 2 \times 0.3086^2 + \frac{2}{3} \times 0.2193^2 + \frac{2}{5} \times (-0.0842)^2 = 0.190468 + 0.032062 + 0.002836.$$

OR

$$\int_{-1}^{+1} P_0(x) dx = \int_{-1}^{+1} 1 dx = 2 \text{ and } \int_{-1}^{+1} P_1(x) dx = \int_{-1}^{+1} P_2(x) dx = 0.$$

+1 point

$$\int_{-1}^{+1} q_2(x)^2 dx = 0.225365 = 0.2254(4dp).$$

OR

$$\int_{-1}^{+1} q_2(x) dx = 2c_0 = 2 \times 0.3086 = 0.6172.$$

6 Linear Algebra

12.0 points · 3 questions

a Use two steps of the Gauss-Seidel method to estimate the solution of the equation

$A\mathbf{x} = \mathbf{b}$, starting at $\mathbf{x}^{(0)}$.

\$\$\$ A = \begin{pmatrix} 5 & 2 & 0 \\ 3 & 6 & -1 \\ -2 & 3 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 2 \\ -4 \\ 1 \end{pmatrix}, \quad \mathbf{x}^{(0)} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} \$\$\$

8.0 points · Open · 1 1/2 Page

+1 point

Gauss-Seidel method $x_i^{(k+1)} = (b_i - \sum_{j < i} A_{ij}x_j^{(i+1)} - \sum_{j > i} A_{ij}x_j^{(i)})/A_{ii}$.

+1 point

$x_1^{(1)} = (b_1 - A_{12}x_2^{(0)} - A_{13}x_3^{(0)})/A_{11} = (2 - 2 \times 0.0000 - 0 \times 3.0000)/5$.

+1 point

$x_2^{(1)} = (b_2 - A_{21}x_1^{(1)} - A_{23}x_3^{(0)})/A_{22} = (-4 - 3 \times 0.4000 - (-1) \times 3.0000)/6 = -2.2000/6$. (Not Jacobi.)

+1 point

$x_3^{(1)} = (b_3 - A_{31}x_1^{(1)} - A_{32}x_2^{(1)})/A_{33} = (1 - (-2) \times 0.4000 - 3 \times (-0.3667))/1$. (Not Jacobi.)

+1 point

$x_1^{(1)} = 2.0000/5 = 0.4000$, $x_2^{(1)} = -2.2000/6 = -0.3667$, $x_3^{(1)} = 2.9000/1$.

+1 point

$x_1^{(2)} = (b_1 - A_{12}x_2^{(1)} - A_{13}x_3^{(1)})/A_{11} = (2 - 2 \times -0.3667 - 0 \times 2.9000)/5$;
 $x_2^{(2)} = (b_2 - A_{21}x_1^{(1)} - A_{23}x_3^{(1)})/A_{22} = (-4 - 3 \times 0.5467 - (-1) \times 2.9)/6$;
 $x_3^{(2)} = (b_3 - A_{31}x_1^{(1)} - A_{32}x_2^{(1)})/A_{33} = (1 - (-2) \times 0.5467 - 3 \times (-0.4567))/1$.

+1 point

$x_1^{(2)} = 2.7333/5 = 0.5467$, $x_2^{(2)} = -2.7400/6 = -0.4567$, $x_3^{(2)} = 3.4633/1$.

+1 point

Reasonable precision (at least 2dp and consistent).

b What is the difference between the *error* and the *residual* of an approximate solution $\tilde{\mathbf{x}}$ to the linear algebraic equation $A\mathbf{x} = \mathbf{b}$?

2.0 points · Open · 1/2 Page

+1 point

Error is norm of difference $\|\mathbf{x} - \tilde{\mathbf{x}}\|$.

+1 point

Residual is (norm of) difference between $A\tilde{\mathbf{x}}$ and $\mathbf{b}$, either $A\tilde{\mathbf{x}} - \mathbf{b}$ or $\|A\tilde{\mathbf{x}} - \mathbf{b}\|$.

c Compute the norm of the residual of your answer to .

2.0 points · Open · 3/5 Page

+1 point

$$\begin{aligned} A\mathbf{x}^{(2)} - \mathbf{b} &= \begin{pmatrix} 5 & 2 & 0 \\ 3 & 6 & -1 \\ -2 & 3 & 1 \end{pmatrix} \begin{pmatrix} 0.5467 \\ -0.4567 \\ 3.4633 \end{pmatrix} - \begin{pmatrix} 2 \\ -4 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 1.8200 \\ -4.5633 \\ 1.0000 \end{pmatrix} - \begin{pmatrix} 2 \\ -4 \\ 1 \end{pmatrix} = \begin{pmatrix} -0.1800 \\ -0.5633 \\ 0.0000 \end{pmatrix}. \end{aligned}$$

+1 point

$$\|A\mathbf{x} - \mathbf{b}\|_{\infty} = 0.5633 = 0.56 \text{ (2sf).}$$

OR

$$\|A\mathbf{x} - \mathbf{b}\|_2 = \sqrt{0.1800^2 + 0.5633^2 + 0.0000^2} = 0.5912 = 0.59 \text{ (2sf).}$$

7 Modelling and Matlab

8.0 points · 2 questions

Consider the following equation for the forced Duffing oscillator:

$$\ddot{x} + \delta \dot{x} + \alpha x + \beta x^3 = A \cos(\omega t)$$

with parameter values

$$\delta = 0.5, \alpha = 0.07, \beta = 0.1, A = 2.3, \omega = 1.9.$$

and initial conditions $x(0) = 2, \dot{x}(0) = 1$.

Text

a Show how to re-write this second-order equation as a system of first-order equations.

4.0 points · Open · 7/10 Page

+1 point

Introduce new variable $v = \dot{x}$, with $\dot{v} = \ddot{x}$.

+1 point

$$\dot{v} = g(t, x, v) = A \cos(\omega t) - (\delta v + \alpha x + \beta x^3).$$

+1 point

State $y = (x, v)$, with $y_1 = x$ and $y_2 = v$.

+1 point

$$\dot{y} = (\dot{x}, \dot{v}) = (v, \ddot{x}) = (v, g(t, x, v)) = (y_2, g(t, y_1, y_2)) =: f(t, y).$$

b Write Matlab code to solve the equation.

4.0 points · Open · 4/5 Page

+1 point

Set parameters, using roman characters:

$\delta=0.5$, $\alpha=0.07$, $\beta=0.1$, $A=2.3$, $\omega=1.9$,

and initial and final time

$t_0=0$, $t_f=\dots$

+1 point

Matlab function syntax:

$f=@(t,y) \dots$

and correct arguments

+1 point

Define right-hand side function:

$f=@(t,y) [y(2); A*\cos(\omega*t) - (\delta*y(2) + \alpha*y(1) + \beta*y(1).^3)]$

OR:

$g=@(t,x,v) A*\cos(\omega*t) - (\delta*v + \alpha*x + \beta*x^3)$, $f=@(t,y)$

$[y(2), g(t, y(1), y(2))]$

and initial condition:

$y_0=[2;1]$

+1 point

Solve using:

$\text{ode45}(f, [t_0, t_f], y_0)$

8 Extra paper

0.0 points · 1 question

Open · 2 9/10 Page